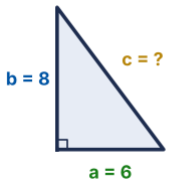
Unit 5 Classwork — Pythagorean Theorem — Answer Key

Hypotenuse · Missing Leg · Converse · Word Problems · Coordinate Distance

A. Find the Hypotenuse

Problem 1 Score 3

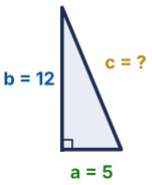
Find the hypotenuse (c) of the right triangle shown.



Answer: 10

Problem 2 Score 3

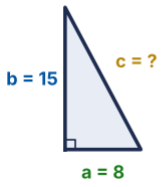
Find the hypotenuse (c) of the right triangle shown.



Answer: 13

Problem 3 Score 4

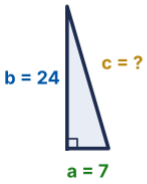
Find the hypotenuse (c) of the right triangle shown.



Answer: 17

Problem 4 Score 4

Find the hypotenuse (c) of the right triangle shown.

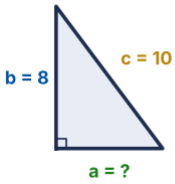


Answer: 25

B. Find a Missing Leg

Problem 1 Score 3

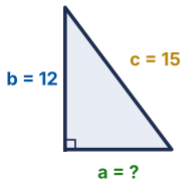
Find the missing leg (a) of the right triangle shown.



Answer: 6

Problem 2 Score 4

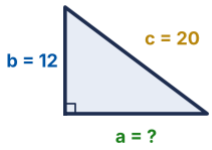
Find the missing leg (a) of the right triangle shown.



Answer: 9

Problem 3 Score 4

Find the missing leg (a) of the right triangle shown.

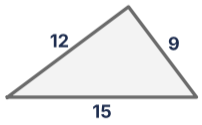


Answer: 16

C. Converse — Is It a Right Triangle?

Problem 1 Score 3

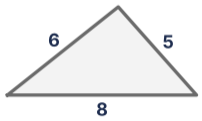
The triangle shown has sides 9, 12, 15. Is it a right triangle?



Answer: yes

Problem 2 Score 4

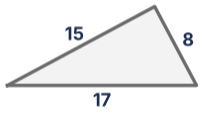
The triangle shown has sides 5, 6, 8. Is it a right triangle?



Answer: no

Problem 3 Score 4

The triangle shown has sides 8, 15, 17. Is it a right triangle?



Answer: yes

Problem 4 Score 5

A triangle has sides 9, 12 and 16. Is it a right triangle? Pick the best reason.

A. No. $9^2 + 12^2 = 225$, but $16^2 = 256$.

B. Yes. 9 + 12 is more than 16.

C. Yes. 9, 12 and 15 make a right triangle, and 16 is close to 15.

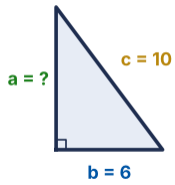
D. No. $9^2 + 16^2$ does not equal 12^2.

Answer: A

D. Word Problems

Problem 1 Score 4

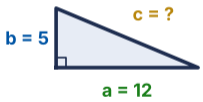
A 10-foot ladder leans against a wall. The base is 6 feet from the wall. How high up the wall does the ladder reach?



Answer: 8

Problem 2 Score 5

A rectangular garden is 12 m long and 5 m wide. What is the length of the diagonal?



Answer: 13

Problem 3 Score 5

A box is 3 cm wide, 4 cm long and 12 cm tall. Find the length of the longest diagonal inside the box, in cm. (Find the diagonal of the bottom first.)

Answer: 13

Problem 4 Score 5

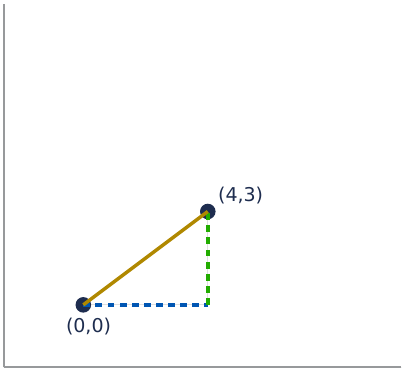
A 13 ft ladder leans on a wall with its base 5 ft from the wall. The base slides out until it is 12 ft from the wall. How many feet does the top of the ladder slide DOWN?

Answer: 7

E. Distance on the Coordinate Plane

Problem 1 Score 4

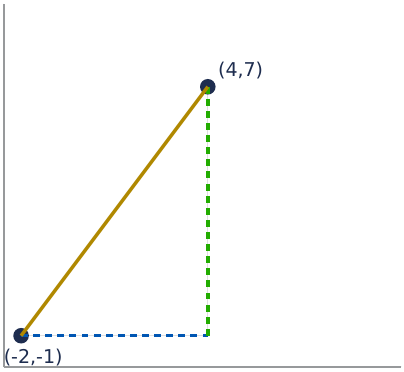
Find the distance between the two points shown.



Answer: 5

Problem 2 Score 5

Find the distance between the two points shown.

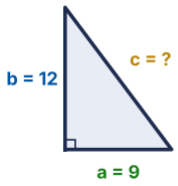


Answer: 10

F. Mixed Review

Problem 1 Score 4

Find the hypotenuse (c) of the right triangle shown.



Answer: 15

Problem 2 Score 4

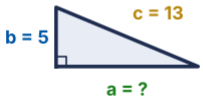
The triangle shown has sides 7, 24, 25. Is it a right triangle?



Answer: yes

Problem 3 Score 5

Find the missing leg (a) of the right triangle shown.



Answer: 12